MILITARY SPECIFICATION

NOZZLE, AUTOMATIC, JET HYPODERMIC INJECTION APPARATUS

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of jet injection nozzle, for use in the administration of intradermal injections.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
QQ-S-763	Steel Bars, Shapes and Forgings, Corrosion-Resisting.
MMM-A-250	Adhesive, Water Resistant, (for Sealing Fiberboard Boxes).
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).

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Military

MIL-A-8625 Anodic Coatings, For Aluminum and Aluminum Alloys.

STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)
24122 NOZZLE, INTRADERMAL, HYPODERMIC JET INJECTION APPARATUS

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply:

AMERICAN NATIONAL STANDARDS INSTITUTE (A. N. S. I.)
A. N. S. I. B46.1 Surface Texture (Surface Roughness, Waviness and Lay)

(Application for copies should be addressed to the American National Standards Institute, 1420 Broadway, New York, New York 10018).

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3. REQUIREMENTS

3.1 Material. All materials used in the fabrication of the nozzle assembly shall be of the highest quality throughout and suitable for the purpose intended. All metal components fabricated from austenitic or martensitic corrosion-resisting steel shall meet the corrosion requirements for class 3 or class 4 material (as applicable) of GG-I-526, in addition to meeting the physical requirements specified in the following paragraphs:

3.1.1 Orifice mount, orifice mount collar, bushing and securing collar. The orifice mount (part 1 on drawing), orifice mount collar (part 3), bushing (part 4) and securing collar (part 5) shall be fabricated from class 303 rod in strict accordance with QQ-S-763. Corrosion requirements shall be as specified in 3.1.

3.1.2 Jewel orifice. The jewel orifice (part 2) shall be fabricated from hexagonal, single crystal structure, 100% aluminum oxide (industrial sapphire). The material shall be chemically inert. The following physical properties shall apply for the finished sapphire:

Melting point	2040^0 Centigrade
Hardness	Knoop: 1525 to 2000; Mohs: 9
Specific Gravity	3.98
Porosity	0%
Compressive strength	300,000 psi at 77^0 Fahrenheit:
Young's Modulus of Elasticity	50 to 55 $X10^6$; psi (Dependent on position of crystal C - axis)
Modulus of Rupture	30^0 Centigrade: 40 to 130,000 psi 540^0 Centigrade: 23,000 to 50,000 psi

It should be noted that the physical properties specified above are minimum standards for the jewel material specified. These properties are not intended to completely describe industrial sapphire as manufactured commercially but only those physical properties that are deemed minimum requirements to provide safety of operation in intended use of nozzle unit.

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3.1.3 Socket-head set screw. The socket-head set screw (part 6) shall be fabricated from austenitic or martensitic corrosion-resistant steel. Corrosion requirements shall be as specified in 3.1.

3.1.4 Protective cap. The protective nozzle cap (part 7) shall be fabricated from a suitable aluminum alloy, 0.019 inch thickness. The cap shall be finished in a black anodized finish. The anodized finish shall be heavy, uniform, continuous black anodic film, free from breaks, scratches and other damage. The exterior of the cap shall be given a grained satin finish prior to anodizing. Immediately after anodizing, and before handling, the cap shall be waxed to protect the finish. Anodic coating shall comply with MIL-A-8625, type II, black-dyed coating.

3.2 Style, design and dimensions. Shall be a jet injection nozzle suitable for use as an accessory with FSN 6515-656-1021 HYPODERMIC INJECTION APPARATUS, JET, AUTOMATIC, 110 Volt, 60 Cycle, AC and FSN 6515-910-0097 HYPODERMIC INJECTION APPARATUS, JET, AUTOMATIC, Foot Operated. (All in accordance with Defense Medical Purchase Descriptions of stated FSN's in effect on the date of invitation for bids). The nozzle shall be suitable for the administration of intradermal injections, such as smallpox, when attached to the above mentioned Federal Stock Numbers. The nozzle shall emit an angular stream which in conjunction with a contoured face, shall effectively introduce the injected agent tangentionally into the skin at the intradermal depth regardless of skin variations due to characteristics such as age, sex, and color. The nozzle shall be fabricated to permit orientation of the angular stream in any desired direction. The injection shall produce a pronounced wheal and shall be characterized by a minimum of bleeding and skin trauma. The nozzle proper shall consist of three sections: bushing, nozzle mount assembly, and securing collar. The nozzle mount assembly shall contain the sapphire jewel orifice (part 2), the orifice mount (part 1), and the orifice mount collar (part 3). The orifice mount collar shall be secured to the orifice mount with a socket-head (Allen type) set screw (part 6). The construction of the nozzle mount shall be such that when the orifice mount collar (part 3) is secured to the orifice mount (part 1), the socket-head set screw shall always be located directly opposite the sapphire jewel orifice (part 2). This orientation of the socket-head set screw to the orifice shall be used when installing the

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nozzle on the injectors specified above. Each nozzle assembly shall be provided with a protective cap (see 3.1.4). The style, design and dimensions of the complete nozzle assembly, including protective cap, shall be in strict accordance with DPSC Drawing 24122.

3.2.1 Jewel orifice and orifice mount. The sapphire jewel shall conform to the dimensions shown on the drawing for part 2, when the jewel contour is checked by contour projection at 100 magnifications as specified in 4.3.2. The faces of the orifice mount shall have a surface finish equal to, or superior to, 32 microinches (MU) in accordance with A. N. S. I Standard B46.1. The orifice of the sapphire jewel shall be perfectly clean and free from lapping or polishing materials. The orifice opening of jewel shall be regular and circular in contour and shall be free from chips or other imperfections. Upon firing of the nozzle when mounted to either FSN 6515-656-1021 or FSN 6515-910-0097, there shall be no leakage of fluid at the mounting junction of jewel and mount and there shall be no side spray in the stream emitted from the nozzle when tested as specified in 4.4.2.

3.2.2 Bushing and securing collar. The bushing (part 4) shall permit attachment of the nozzle mount assembly (as described in 3.2 and shown on drawing) to FSN 6515-656-1021 and FSN 6515-910-0097 by means of the securing collar (part 5).

3.3 Interchangeability. The nozzle assembly shall be designed and manufactured so that all removable component parts shall be interchangeable with each other and all standard automatic hypodermic injection apparatus specified herein.

3.4 Cleaning and passivation. Metal parts, after fabrication, shall be cleaned in accordance with good commercial practice. After machining, all corrosion-resisting steel parts shall be ultrasonically cleaned and passivated. After assembly of jewel orifice into orifice mount, the nozzle mount assembly (see 3.2) shall again be ultrasonically cleaned and passivated.

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3.5 Clinical tests. When deemed necessary (if contractor has not previously supplied acceptable injectors), the Government shall arrange for the clinical testing of the intradermal nozzles as specified in 4.4.1.

3.6 Service data. The contractor shall enclose, with each nozzle, one instruction sheet that contains complete instructions for installation, operation, maintenance, and a list of component parts. The nozzle shall be identified in the service data and parts lists by the manufacturer's part number. As soon as practical after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed service data, for review and comment by the Defense Personnel Support Center, Directorate of Medical Materiel. The cover of each instruction sheet shall indicate the following:

FSN 6515-913-7908
NOZZLE, AUTOMATIC JET HYPODERMIC INJECTION APPARATUS
FOR: FSN 6515-656-1021 and FSN 6515-910-0097
HYPODERMIC INJECTION APPARATUS, JET AUTOMATIC
Contractor's Trademark, Model Number, Name and Address.

In order to minimize accidental striking of nearby personnel when hypodermic injection jet apparatus is primed or fired into the air (after the nozzle is assembled to injection apparatus), warning shall be made in service data to point nozzle orifice in a downward direction (i.e. in the same direction of apparatus trigger) when the nozzle is assembled to injection apparatus. The preliminary copies of the instruction sheet shall be submitted sufficiently early to permit adequate review by the Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of required revisions and corrections in the final instruction sheet without delaying delivery under the contract or purchase order. In

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addition, the contractor shall mail 15 copies of the approved service data to the contracting officer for distribution. Should the supplier have furnished acceptable instruction sheets for the identical item within two years preceding the date of contract, and propose to furnish instruction sheets identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies of the approved service data may be waived upon request.

3.7 Workmanship. Workmanship shall be first class throughout. Nozzle shall be free from defects which detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. Certificates of quality supplied by the manufacturer of the corrosion resisting metals and aluminum oxide may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, the individual test results, and the specification requirements.

4.2 Sampling.

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4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in Table I. Unit of product for examination shall be one nozzle.

Table I. Sampling Criteria For Examination

Type Defect	Inspection Level *	AQL (percent defective)
Major	S-4	1.0
Minor	S-2	2.5

* If acceptance number is not exceeded, lot shall be accepted. If rejection number is equalled or exceeded, then the inspection at level II may be performed to determine the final rejection or acceptance.

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table II. Sampling and examination of finished nozzle performance tests shall be on a 100% basis. Unit of product for test purposes shall be one nozzle.

Table II. Sampling Criteria For Tests

Component	Characteristic	Req.	Test Procedure	Insp. Level	AQL
Nozzle	Boil Test	3.1	GG-I-526b	S-2	*
	Chemical Composition	3.1.1	QQ-S-763	QQ-S-763	*
	Mechanical Properties	3.1.1	QQ-S-763	QQ-S-763	*
	Dimensions	3.2	----	S-2	*
	Performance Tests	3.2.1	4.4.2	100%	*
	Clinical Tests (when necessary)	3.5	4.4.1	S-4	*

* Acceptance number shall be zero for all sample sizes.

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4.2.3 Sample unit usage. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The nozzles shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in Tables I and III. Inspection, however, shall not be restricted to defects listed in Table III. Any nozzles in the sample containing one or more defects shall be rejected and if the number of defective items in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

Table III. Classification of Defects

Categories	Defects
Critical	None defined.
Major	
101	Protective cap not furnished.
102	Finish of orifice mount not as specified.
103	Nozzle not free from cracks, burrs, scale, pits, sharp edges and tool marks.
104	Orifice of nozzle not free from lapping or polishing compound, or other foreign material.
105	Set screw not assembled to collar orifice mount.
106	Service data not furnished.
Minor	
201	Cover of service data not as specified.
202	Unit package marking not as specified.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

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4.4.1 Clinical tests. The government will arrange for the clinical testing of the nozzles on patients to determine the proper functioning of the unit, including administration of typical intradermal injections to insure production of proper wheals with a minimum of bleeding. A standard hypodermic jet injection apparatus (FSN 6515-656-1021 or 6515-910-0097) shall be used in giving injections.

4.4.2 Nozzle performance tests. The sapphire jewel orifice contours shall be checked by means of contour projection at 100 magnifications (using diiodomethane) to determine dimensional compliance with the drawing. After mounting of the sapphire, the nozzle shall be placed on a standard hypodermic jet injection apparatus (FSN 6515-656-1021 or 6515-910-0097) and the injector fired a minimum of 25 times. The stream emitted shall be free from all side spray. There shall be no trace of fluid leakage around the periphery of the orifice insert or any loosening of the orifice insert in the mount. Each jet nozzle assembly shall be tested as described above.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One nozzle, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	144 units

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5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be over-packed in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each nozzle shall be inserted into a clear cellulose acetate container with a natural polyethylene cover. Size of the container shall be 1-1/4 inch diameter by .018 wall be 1-1/8 inch overall legnth. Closure shall be adequate to prevent accidental opening.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2-inch wide tape conforming to PPP-T-76. Closing of bottom flap with adhesive conforming to MMM-A-250 or metal stitches is permitted. In addition, twelve instruction sheets shall be placed within the container.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B.

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5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type ; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, constructed so as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown. In addition, the intermediate container shall be marked with the legend:

"Enclosed instruction sheet shall be furnished with each unit package."

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The jet injection nozzles are to be used in the administration of intradermal injections (such as smallpox). Shall be suitable for use with FSN 6515-656-1021 and FSN 6515-910-0097.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Applicable levels of packing and packaging.
c. Federal stock number.

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6.3 This specification covers the following item listed in the Federal Supply Catalog:

FSN 6515-913-7908 NOZZLE, AUTOMATIC, JET HYPODERMIC INJECTION APPARATUS.

6.4 Supersession data. This specification includes requirements of Purchase Description No. 1, dated 20 February 1968.

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing activity:

Defense Supply Agency - DM
Project No. 6515-1450

Review activities:

Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-N-36667 FSN:

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 68 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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